

Introduction

- ❖ Urban areas face challenges with dry leaf accumulation, causing drainage blockages, fire hazards, and hygiene issues.
- ❖ Manual tools like rakes and blowers are labor-intensive, slow, and inefficient for large spaces.
- ❖ Automatic machines collect, transfer, and optionally shred leaves, reducing effort and improving efficiency.
- ❖ Shredded leaves are easier to handle and can be reused for composting, supporting sustainable waste management.

Purposes of Machine

- ❖ To collect the dry leaves
- ❖ To reduce the risk of harm to people
- ❖ To reduce manual labour
- ❖ To increase efficiency
- ❖ To reduce time and effort
- ❖ To be Eco -Friendly operation
- ❖ To improve hygiene and Reduce Air Pollution

Working Procedure of Machine

1. **Collection** – Rotating brushes or vacuum suction lifts leaves from the ground.
2. **Transfer** – Blower, conveyor, or duct moves leaves toward storage.
3. **Storage/Processing** – Leaves are stored, compacted, or shredded for easier handling and composting.
4. **Automation** – Advanced models use sensors for navigation, dust suppression, and efficiency.



Advantages

1. Eco-Friendly Operation
2. Low Noise Output
3. User-Friendly Design
4. Safe to Use
5. Low Maintenance Requirements
6. Cost-Efficient in the Long Term
7. No Fuel Storage Needed
8. Instant Start and Stop
9. Ideal for Urban and Institutional
10. Reduces Manual Labor Time



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“Automatic Dry Leaf Collecting Machine”



Classification of Machine

The following classification criteria help categorize automatic dry leaf collecting machines based on their physical structure, operating mechanism, performance, and intended applications.

1. Mobility Configuration
2. Collection Mechanism
3. Power Source
4. Collection Capacity
5. Operating Mode

Types of Dry Leaf Machine

- ❖ Leaf Shredder / Mulcher – Cuts leaves into small pieces for compost
- ❖ Leaf Collector / Vacuum – Sucks and stores dry leaves
- ❖ Leaf Baler – Compresses leaves into bales
- ❖ Leaf Crusher – Turns leaves into fine powder or particles

Components of Dry Leaf Machine

1. Battery
2. DC Motor
3. Shaft
4. Pulley
5. Wheels
6. Trash Bin
7. Belt
8. Controller
9. Main frame
10. Ball bearings
11. Leaf collector brush

